Hongming Xu

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GitHub ◇ Webpage

Research Interests

My research interests lie in **agentic AI and foundation models**. I focus on **agentic reinforcement learning, model adaptation, and agent memory**, particularly how agents learn from interaction feedback, adapt model parameters, and accumulate and reuse experience over long horizons. My background in **database systems and data management** also informs my interest in organizing, retrieving, and maintaining agent memory.

Education

Shanghai Jiao Tong University — Shanghai, China
Undergraduate in School of Computer Science — Sept. 2024 – Present
Member of the Zhiyuan Honors Program

Selected Publications & Manuscripts

MemTrace: State-Consistent Memory for Long-Horizon Coding Agents

Under review at ICLR 2027 | **First author** **[arXiv]** **[Repository]** — July 2026 – Present

Developed a state-consistent memory system that combines structured execution traces with repository grounding to validate historical evidence before reuse. Evaluated across three software-engineering benchmarks and two coding-agent harnesses, consistently improving the primary metric over native memory strategies.

Are We Ready for An Agent-Native Memory System?

Under review at VLDB **[arXiv]** **[Code]** — Mar. 2026 – June 2026

Contributed to a systematic study of agent memory across representation and storage, extraction, retrieval/routing, and maintenance. Evaluated task effectiveness, retrieval fidelity, dynamic-update robustness, long-term stability, and runtime cost across diverse workloads.

Dial: A Knowledge-Grounded Dialect-Specific NL2SQL System

PVLDB 2026 | **Co-first author (listed second)** **[arXiv]** **[Code]** — Aug. 2025 – Mar. 2026

Contributed to natural-language preprocessing, dialect-aware logical planning, and knowledge retrieval for multi-dialect NL2SQL. Helped design and initialize **HINT-KB**, prepare datasets, and develop iterative feedback and semantic correction; participated in failure analysis and rebuttal revisions.

Research Experience

MemTensor

Research Intern — Mentor: Juncheng Zhang (Tsinghua University) **[Technical Report]** — Aug. 2026 – Present

- Contribute to post-training of the **284B-parameter DeepSeek-V4-Flash-0731**. Designed a **DeepSWE** optimization strategy by synthesizing **550 validated** staged, ablation, and repair tasks from 113 original SWE tasks, and optimizing difficulty-aware data mixing, improving DeepSWE performance by nearly **10%**.
- Explore memory mechanisms for long-horizon coding agents, focusing on persistent execution history, repository-state alignment, and reliable reuse of historical evidence; led the first-author **MemTrace** project on state-consistent memory for repository-scale coding tasks.

Shanghai Jiao Tong University — Theseus Lab

Undergraduate Researcher — Supervisor: Prof. Xuanhe Zhou — Aug. 2025 – June 2026

- Worked on **Dial**, a knowledge-grounded dialect-specific NL2SQL system. Contributed to dialect-aware logical planning and the construction of **HINT-KB**, including knowledge organization, initialization, retrieval components, semantic correction, and failure analysis for heterogeneous SQL dialects.
- Extended this data-management perspective toward **agent memory and memory management** through *Are We Ready for An Agent-Native Memory System?*. Studied memory representation and storage, extraction, retrieval, maintenance, and systematic evaluation, forming the basis for my later work on long-horizon coding-agent memory.

Honors

Zhiyuan Honors Scholarship

Personal Interests

Badminton and fitness.